(Please write your Enrollment Number) Enrollment No. 02501022023

End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name:<B.Tech. >
(For students admitted on and after 2022)
Semester:<2nd> 2023-24

Subject Code: BAS 102	Subject: Applied Mathematics II
Time :3 Hours	Maximum Marks :60
Note:Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)	
	(a) Find the directional derivatives of $\emptyset = xy^2 + yz^2$ at the point $(2,-1,1)$ in the direction of the vector $\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.		
	(b)Evaluate $\int_C \vec{F}.d\vec{r}$ along the parabola $y^2 = x$ between the points (0,0) and (1,1) where$\vec{F} = x^2\hat{\imath} + xy\hat{\jmath}$.		
	(c) Solve $\frac{d^3y}{dx^3} + y = 0$.		
	(d) Find the particular integral of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$.		
	(e) Find the Laplace transform of the function $t^2 \cos at$.		
	(f)Evaluate $L\left(\int_0^t \frac{\sin t}{t} dt\right)$.		
	(g)Evaluate $\int_C (2\bar{z} - z)dz$, where C is $x = -t,\ y = t^2 + 2,\ 0 \le t \le 2$.		
	(h)Expand $f(z) = \sin z$ in a Taylor's series about $z = \frac{\pi}{4}$ and determine the radius of convergence of this series.		
	UNIT-I		
Q2	(a) Verify Stoke's theorem for the vector field $\vec{F} = (2x - y)\hat{\imath} - yz^2\hat{\jmath} - y^2z\hat{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$, bounded by its projection on xy-plane. (b) Evaluate $\iint_S \vec{F}.\hat{n}dS$ where $\vec{F} = 3y\hat{\imath} + 2z\hat{\jmath} + x^2yz\hat{k}$ and S is the surface $y^2 = 5x$ in the positive octant bounded by the planes $x =$ 3and z= 4.	(5*2=10)	
Q3	(a) Using Stoke's theorem, evaluate $\int_C [(x + y)dx + (2x - z)dy + (y + z)dz]$, where C is the boundary of the triangle with vertices $(2,0,0), (0,3,0)$ and $(0,0,6)$. (b) Evaluate $\iint_S \vec{F}.\hat{n}dS$ using Gauss Divergence theorem where	(5*2=10)	

	$\vec{F} = 2xy\hat{i} + yz^2\hat{j} + zx\hat{k}$ and S is the surface of the region bounded by $x= 0, y = 0,\ z = 0, y = 3, x + 2z = 6.$	
UNIT-II		
Q4	(a) Solve $y'' - 2y' + 2y = e^x \tan x$ by method of variation of parameters. (b)Solve $$(2x+3)^2\frac{d^2y}{dx^2} - (2x+3)\frac{dy}{dx} - 12y = 6x;$$ $$x \neq -\frac{3}{2}.$$	(5*2=10)
Q5	(a) Solve the differential equation $y'' - 4y' + 4y = 5e^{2x}$ using method of undetermined coefficients. (b) Solve the simultaneous equations $$(D+1)x + (2D+1)y = e^t$$ $$(D-1)x + (D+1)y = 1$$	(5*2=10)
UNIT-III		
Q6	(a) Derive the formula for $L[f'(t)]$, if $L[f(t)] = F(s)$. (b) Using Laplace transform, solve $\frac{d^2y}{dt^2} + \frac{dy}{dt} = t^2 + 2t$ where $y = 4$ and $y' = -2$ when $t = 0$.	(5*2=10)
Q7	(a)Find $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$ using Convolution theorem. (b)Solve $\frac{dy}{dt} + 2y + \int_0^t y\,dt = \sin t,\ y(0) = 1.$	(5*2=10)
UNIT-IV		
Q8	(a) Prove that for the function $f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}$, $z \neq 0$, and $f(0) = 0$, the Cauchy's Riemann equations are satisfied at the origin, but $f'(0)$ does not exist. (b) Evaluate $\oint_C \frac{3z^2+z}{z^2-1}dz$, where C is the circle $$\lvert z - 1\rvert = 1.$$	(5*2=10)
Q9	(a) Expand $f(z) = \frac{1}{(z-1)^2(z-3)}$ in a Laurent series valid for $0 < \lvert z - 1\rvert < 2$. (b)Using the residue theorem, evaluate $\oint_C \frac{dz}{z^3(z+4)}$, where C is the circle $\lvert z + 2\rvert = 3$.	(5*2=10)